

Name: _____

Date: _____

VECTORS, (WRITING AND REPRESENTING VECTORS)

LO: I can write vectors using column notation and represent them on a grid.

Writing and representing vectors

A **vector** describes a movement with both magnitude (size) and direction. We write column vectors as **(x, y)** where x is the horizontal movement and y is the vertical movement.

Convention: Positive x means right, negative x means left. Positive y means up, negative y means down. Vectors are equal if they have the same magnitude and direction.

Quick examples

●	(3, 2) means 3 right and 2 up	positive components
●	(-1, 4) means 1 left and 4 up	negative x means left
●	(3, 2) and (3, 2) drawn at different points are equal	same magnitude and direction
●	(3, 2) means 3 up and 2 right	x is horizontal (first), y is vertical (second)

Key idea

A column vector gives horizontal then vertical displacement. Equal vectors have the same column vector regardless of starting point.

vector = **(horizontal, vertical)**

Right and up are positive; left and down are negative

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - writing a column vector

Write the vector that moves from A(1, 3) to B(4, 7) as a column vector.

Horizontal: $4 - 1 = 3$ (right)

Vertical: $7 - 3 = 4$ (up)

Column vector: (3, 4)

(3, 4)

Subtract the start coordinates from the end coordinates.

Example 2 - negative components

Write the vector from P(5, 6) to Q(2, 1).

Horizontal: $2 - 5 = -3$ (left)

Vertical: $1 - 6 = -5$ (down)

Column vector: (-3, -5)

(-3, -5)

Negative values show movement left or down.

Example 3 - drawing a vector on a grid

Draw the vector (4, -2) starting from point (1, 5).

Start at (1, 5)

Move 4 right: $x = 1 + 4 = 5$

Move 2 down: $y = 5 + (-2) = 3$. End at (5, 3)

Arrow from (1, 5) to (5, 3)

Plot the start, apply the vector, mark the end with an arrow.

Example 4 - identifying equal vectors

Are (2, 3) from A(0, 0) and (2, 3) from B(4, 1) equal vectors?

First vector: same column (2, 3)

Second vector: same column (2, 3)

Same magnitude and direction, so they are equal

Yes, they are equal

Position does not matter; only magnitude and direction define a vector.

YOUR TURN – PRACTICE (deliberate practice)

1. Writing column vectors

Write each movement as a column vector.

- a) 3 right, 5 up
- b) 2 left, 4 up
- c) 6 right, 1 down
- d) 4 left, 3 down

2. From coordinates

Find the column vector from the first point to the second.

- a) (1, 2) to (4, 6)
- b) (5, 7) to (2, 3)
- c) (0, 0) to (6, -2)
- d) (3, 8) to (3, 1)

3. Negative and zero components

Write the column vector.

- a) (7, 4) to (7, 9)
- b) (3, 5) to (-2, 5)
- c) (-1, -3) to (2, -6)
- d) (-4, 2) to (-1, -4)

4. Drawing vectors

Starting from the given point, state the end point after applying the vector.

- a) Start (2, 3), vector (4, 1)
- b) Start (5, 6), vector (-3, 2)
- c) Start (0, 4), vector (2, -4)
- d) Start (7, 1), vector (-5, -1)

5. Mixed vector problems

Solve each problem.

- a) Write the return vector (opposite) of (3, -2)
- b) A to B is (4, 5). B to C is (1, -3). Find A to C.
- c) P to Q is (-2, 6). What is Q to P?
- d) Vector (a, 3) moves (2, 1) to (5, 4). Find a.
- e) Are (4, -1) and (-4, 1) equal, opposite, or neither?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) (3, -2) means 3 right and 2 down ☐

Correct answer: _____

Reason: _____

b) (3, -2) means 3 up and 2 left ☐

Correct answer: _____

Reason: _____

c) Two vectors with the same column vector are equal regardless of position ☐

Correct answer: _____

Reason: _____

d) (0, 5) means no movement at all ☐

Correct answer: _____

Reason: _____

e) The negative of (a, b) is (-a, -b) ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A drone starts at position (2, 3) on a grid. It follows three vectors in sequence: (4, 1), then (-2, 5), then (1, -3). What is the drone's final position? Write a single vector that would take the drone directly from start to finish.

Expression: _____

Simplified: _____

b) Points A(1, 2), B(4, 6), C(7, 10) lie on a grid. Write the vectors AB and BC as column vectors. What do you notice? Explain why A, B, and C must be collinear (lie on the same straight line).

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

